

67.pdf



University of Engineering & Management

Document Details

Submission ID

trn:oid:::3618:103124128

Submission Date

Jul 1, 2025, 2:33 PM GMT+5:30

Download Date

Jul 5, 2025, 12:04 AM GMT+5:30

File Name

67.pdf

File Size

786.5 KB

12 Pages**3,777 Words****21,977 Characters**

30% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **76 Not Cited or Quoted 28%**
Matches with neither in-text citation nor quotation marks
-  **5 Missing Quotations 1%**
Matches that are still very similar to source material
-  **1 Missing Citation 1%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 20%  Internet sources
- 22%  Publications
- 24%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 76 Not Cited or Quoted 28%**
Matches with neither in-text citation nor quotation marks
- 5 Missing Quotations 1%**
Matches that are still very similar to source material
- 1 Missing Citation 1%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 20% Internet sources
- 22% Publications
- 24% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Publication	M. Raveenthini, R. Lavanya, Raul Benitez. "Grad-CAM based explanations for mult...	2%
2	Internet	export.arxiv.org	2%
3	Internet	www.mdpi.com	1%
4	Submitted works	Heriot-Watt University on 2023-11-28	1%
5	Submitted works	The British College on 2025-01-05	1%
6	Submitted works	University of Nottingham on 2025-04-18	1%
7	Submitted works	Macao Polytechnic Institute on 2024-03-29	1%
8	Internet	www.science.org	1%
9	Publication	Arpana Prasad, V. Asha, Neha, Naveen Kumar P, Imran Pasha, Junaid Pasha M. "...	<1%
10	Internet	www.ripublication.com	<1%

11	Submitted works	University of Wales Swansea on 2024-11-25	<1%
12	Submitted works	VIT University on 2025-04-21	<1%
13	Internet	etasr.com	<1%
14	Submitted works	University of Warwick on 2021-09-20	<1%
15	Internet	jase.tku.edu.tw	<1%
16	Publication	"Future Data and Security Engineering. Big Data, Security and Privacy, Smart City ...	<1%
17	Submitted works	Tufts University on 2025-03-06	<1%
18	Internet	dtai.tsue.uz	<1%
19	Publication	Arvind Dagur, Karan Singh, Pawan Singh Mehra, Dharendra Kumar Shukla. "Artific...	<1%
20	Publication	V. Sharmila, S. Kannadhasan, A. Rajiv Kannan, P. Sivakumar, V. Vennila. "Challeng...	<1%
21	Internet	ri.conicet.gov.ar	<1%
22	Submitted works	Universiti Teknologi Malaysia on 2023-10-04	<1%
23	Internet	www.isteonline.in	<1%
24	Publication	"Fifth Congress on Intelligent Systems", Springer Science and Business Media LLC...	<1%

25	Submitted works	University of Lancaster on 2024-05-26	<1%
26	Internet	theses.liacs.nl	<1%
27	Submitted works	Addis Ababa University on 2023-09-06	<1%
28	Publication	"Advanced Information Networking and Applications", Springer Science and Busi...	<1%
29	Submitted works	George Bush High School on 2023-07-12	<1%
30	Submitted works	Liverpool John Moores University on 2024-12-14	<1%
31	Submitted works	University of Greenwich on 2022-06-29	<1%
32	Publication	"Intelligent Systems Design and Applications", Springer Science and Business Me...	<1%
33	Publication	Dominika Matus, Frederic Runge, Jörg K.H. Franke, Lars Gerne, Michael Uhl, Frank...	<1%
34	Publication	Fernandes, Rendson R.. "Enhancing Contrastive Learning with Soft Prompts for M...	<1%
35	Submitted works	Liverpool John Moores University on 2024-05-13	<1%
36	Submitted works	Munster Technological University (MTU) on 2025-06-11	<1%
37	Submitted works	University of Stirling on 2024-08-30	<1%
38	Publication	Wajahat Akbar, Abdullah Soomro, Mohib Ullah, Muhammad Inam Ul Haq, Sana Ul...	<1%

39	Internet	arxiv.org	<1%
40	Internet	deepstream.datascouting.com	<1%
41	Internet	hal.archives-ouvertes.fr	<1%
42	Internet	peerj.com	<1%
43	Publication	Abdul Rafay, Zaeem Asghar, Hamza Manzoor, Waqar Hussain. "EyeCNN: explorin...	<1%
44	Submitted works	Florida International University on 2023-04-30	<1%
45	Submitted works	Liverpool John Moores University on 2020-11-05	<1%
46	Publication	Sanger, Mario. "Representation Learning for Biomedical Text Mining", Humboldt ...	<1%
47	Submitted works	University of Northumbria at Newcastle on 2021-08-26	<1%
48	Submitted works	University of Queensland on 2024-05-07	<1%
49	Submitted works	University of Rijeka on 2022-01-17	<1%
50	Internet	injoit.org	<1%
51	Internet	irjet.net	<1%
52	Internet	kth.diva-portal.org	<1%

53	Internet	mail.ijain.org	<1%
54	Internet	www.statmt.org	<1%
55	Publication	"Computer Vision and Image Processing", Springer Science and Business Media L...	<1%
56	Submitted works	Eiffel Corporation on 2023-12-07	<1%
57	Submitted works	University of West London on 2025-05-29	<1%
58	Publication	Pishahang, Mohammad. "Wildfire Safe Evacuation Planning and Management", U...	<1%
59	Submitted works	The Robert Gordon University on 2025-02-22	<1%

Eye disease classification using Feature Fusion based Deep models

Priya Majumdar¹, Subarna Das¹, Samagata Sanyal¹, Sonia Thakur¹, Anindita Maity¹, Ritaja Bhattacharya¹, Pritam Goswami¹, and Sourajit Maity^{1,2}

¹ Sister Nivedita University, Newtown, Kolkata, India

² Jadavpur University, Kolkata, India

priyamajumdar606@gmail.com, subarnad2000@gmail.com,
samagatasanyal6289@gmail.com, soniathakur.13613@gmail.com,
maityanindita01@gmail.com, nehabhatta00@gmail.com,
pgoswami.academic@gmail.com sourajit.cse.ju@gmail.com

Abstract. Early diagnosis of ocular diseases such as cataract, diabetic retinopathy, glaucoma, and other vision-impairing diseases is crucial to avoid permanent blindness. Though classification of such diseases using deep learning approaches has made a significant progress but still there is some gap on robust accuracy across various environments that the previously existing works were unable to achieve.

In this paper, we propose a novel approach to classify retinal fundus images into four disease classes. For this a publicly available dataset [1] with 4217 high-resolution images are used and considerable data augmentation methods were employed in order to enhance generalization. We first used the pretrained models such as InceptionV2, EfficientNetB2, Xception, InceptionV3 and MobileNetV2. Among these models the best accuracy was attained in Xception and MobileNetV2 models. Thus, our novel approach is based on the fusion of two feature maps extracted from Xception and MobileNetV2 baseline models, respectively. Additionally, we employ a Convolutional Block Attention Module (CBAM) to enhance the resilience and precision. With this approach we have achieved an accuracy of 98.79%. Class-wise metrics indicated exceptionally high performance in identifying cataract and glaucoma. The effectiveness of the model was also tested using t-SNE visualization, Precision-Recall curves, and a confusion matrix, all of which affirmed good class separability and confidence of predictions. All these findings indicate that the presented model is a promising method for retinal disease classification in an automated, accurate, and interpretable manner, and it has strong potential to be integrated into teleophthalmology platforms.

Keywords: Deep Learning · CBAM · Retinal Disease · Feature Fusion · Eye Health · Fundus Image · Attention Mechanism

1 Introduction

Since the dawn of Artificial Intelligence (AI), it has been applied in various fields for various problems such as medical diagnosis[2], autonomous vehicles[3],

2 Majumdar et al.

natural language processing [4], and predictive analytics [5]. In recent days, AI plays a huge role in medical fields. From medical diagnosis[2] to Surgical robots [6], and even pharmaceuticals [7], AI has come into the limelight due to its immense success in each of these fields.

AI has been immensely used to diagnose diseases early since its beginning. There are many disease that is hard to diagnose but can cause severe physical and mental harm to patients if not diagnosed in time (e.g., many forms of cancer, various mental disorders, some eye related diseases etc.). There has been many works that can detect these kind of diseases early. For example, in[8] and [9] one can find the recent developments in early cancer detection and dignosis of mental health disorder respectively.

In this work, we are interested in detection of Eye diseases with images. Eye disease is the leading cause of blindness in the world and affects millions of people throughout their life. Visual impairment has a much larger influence than health, with mental health, social integration, capacity to work, and general independence also affected. Blindness has the potential to greatly reduce the ability to perform activities of daily living, causing loss of independence, isolation, and a decreased quality of life. One of the biggest issues with eye diseases is that they can start to develop without anyone realizing it. Most conditions will develop symptom-free for years and in that time the disease can progress silently. By the time vision is affected, irreversible damage may have already occurred. This delay makes treatment more difficult and, in severe cases, less effective. Many diseases, such as glaucoma, diabetic retinopathy, macular degeneration, and cataract, develop quietly, and early diagnosis and treatment are necessary to prevent permanent visual impairment and overall quality of life.

There has been a vast literature on the classification of eye related diseases using various approaches and methods. In the following subsection one can find the recent developments.

1.1 Literature Review

In [10], Prasad et al. proposed an approach based on CNN to detect eye diseases. The data they used has two classes Diabetic Retinopathy (DR) and Glaucoma. In 2021, the authors of [11] used deep convolutional neural networks to detect eye diseases early. They have used fundus images with eight different disease classes as their dataset. The survey paper [12] presents us with a detailed overview on the research works until 2022 on eye disease detection using various methodologies such as machine learning, Deep Learning, KNN, SVM, CNN etc. A novel approach based on SVM was introduced by Marouf et al. in their work [13] to detect several eye diseases. The relatively higher accuracy was achieved due to their usage of ranker-based feature selection methods in their work. The data was collected through in-person interviews and has five different classes namely, Cataracts, Acute Angle-Closure Glaucoma, Primary Congenital Glaucoma, Bulging Eyes and Ocular Hypertension. Rahaman and Sait proposed another method based on SSD, WOA with levy flight and wavelet search and ShuffleNetV2 model [14]. They have used the ODIR data set consisting of left and

right eye images and EDC data set. Finally in a very recent work by Sharma et. al. [15], Deep transfer learning model based on EfficientNetB5 is used to classify fundus images into different disease classes.

In this paper, we have proposed a novel approach that can classify a retinal fundus image into a class of four diseases namely, glaucoma, diabetic retinopathy, macular degeneration, and cataract. Primarily we have used pretrained models such as InceptionV2 [16], EfficientNetB2 [17], Xception [18], InceptionV3 [19] and MobileNetV2 [20] on the dataset [1] where we achieved accuracy of 85.78%, 90.33%, 91.66%, 90.82% and 91.66% respectively. We have noticed that the best accuracy was achieved by using Xception and MobileNetV2 models. Thus we design our model as a fusion of two feature maps extracted from Xception and MobileNetV2 respectively. Further, we employ a Convolutional Block Attention Module (CBAM) to enhance the precision. This approach has led us with an accuracy of 98.79%.

2 Methodology

In this research, we utilized several state-of-the-art deep learning models—InceptionV3, Xception, InceptionV2, MobileNetV2, and EfficientNetB2 for retinal disease classification. Among them, MobileNetV2 and Xception achieved the best performance in terms of training and validation accuracy, making it the most effective model for efficient and accurate classification of retinal images.

2.1 MOBILENETV2

MobileNetV2 implements inverted residual blocks with linear bottlenecks to enhance efficiency and accuracy. As shown in Fig. 1, the network expands the input data before performing depthwise convolutions and then reduces it to a lower-dimensional output. Linear bottleneck helps preserve information. The design provides exceptional efficiency for mobile devices while delivering better speed and performance than its predecessor.

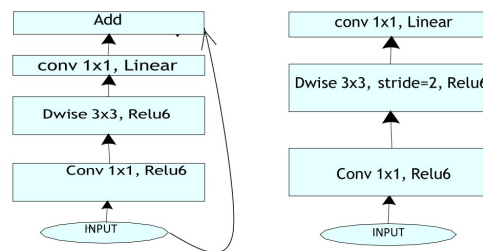


Fig. 1. Architecture of MobileNetV2

4 Majumdar et al.

2.2 XCEPTION

Xception (Extreme Inception) is a deep convolutional neural network architecture that is employed as the backbone for image classification. As shown in the Fig. ??, the model utilizes depth wise separable convolutions, which decompose the standard convolution operation into a depth wise and pointwise convolution, thereby reducing computational complexity while preserving model performance. This architecture allows for efficient learning of spatial and channel-wise features making it well suited for high dimensional image data.

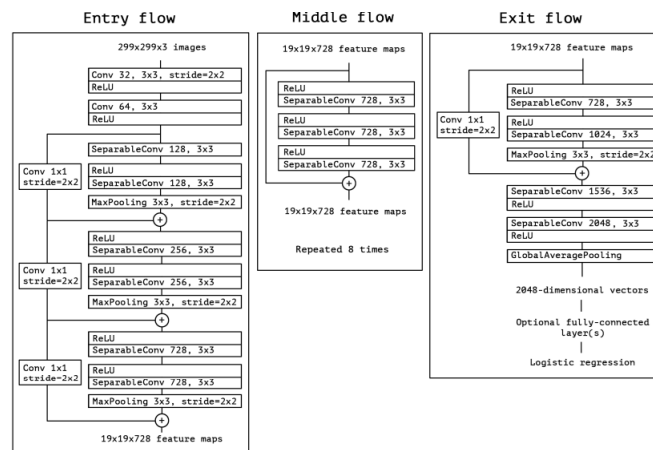


Fig. 2. Architecture of Xception [18]

2.3 Deep Feature Fusion

It fuses two feature maps that are extracted from two different base models MobileNetV2 and Xception in two sequential paths.

2.4 CBAM

The Convolutional Block Attention Module (CBAM) is a light-weight attention module that boosts CNN performance by sequentially applying channel and spatial attention (See Fig. 3).

It learns where and what to attend to by highlighting relevant features and diminishing less important ones. CBAM can be directly incorporated into the existing CNNs with zero overhead and enhances performance and interpretability in both classification and detection tasks.

Eye disease classification using Feature Fusion based Deep models

5

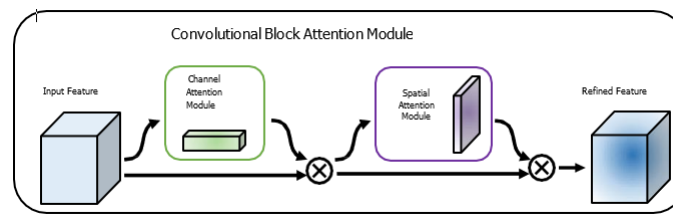


Fig. 3. Architecture of CBAM

2.5 Channel Attention Module

The Channel Attention Module attends to "what" is significant in a feature map by modeling inter-channel dependencies. Both average pooling and max pooling are utilized to produce descriptors, which are then passed through a shared MLP to yield a 1D channel attention map. The map is then used to weigh the input feature to highlight significant channels.

2.6 Spatial Attention Module

The Spatial Attention Module targets "where" in the feature map matters by extracting spatial dependencies. It uses average and max pooling between channels to create a combined descriptor, which is subsequently passed through a convolution layer to generate a 2D spatial attention map. The map serves to emphasize significant areas in the feature space.

2.7 Multi-Head Attention

Multi-Head Attention enables a model to focus on various representation subspaces using multiple attention heads, each having its own key, query, and value projections. Although this enhances expressiveness in models, research indicates that most heads learn redundant features, which makes it feasible to compress parameters using shared projections. Collaborative attention resolves this by allowing heads to learn shared key/query spaces while preserving performance while minimizing redundancy and model size.

2.8 Proposed Architecture

As shown in Fig. 4, for the proposed architecture in our work, an input-tensor of shape (400,400,3) is created at the beginning and then it is passed parallelly through two base models, namely MobileNetV2 and Xception, pre-trained with the ImageNet weights without including the top fully connected layers of both models. The feature maps that are extracted are then subsequently navigated through the GlobalMaxPooling2D (GMP) layer and a dropout layer to get the feature maps.

6 Majumdar et al.

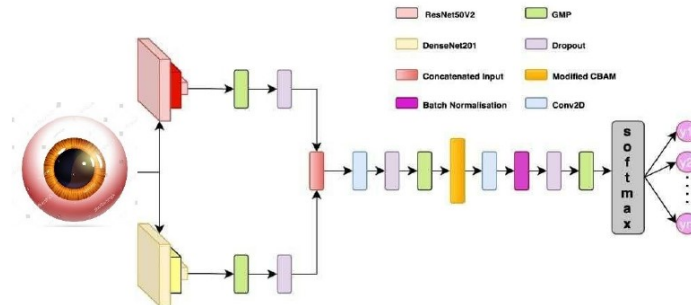


Fig. 4. The architecture of our proposed MobileNetV2-Xception with CBAM approach

The feature maps that we got are then fused using simple concatenation. As a result we have achieved a combined feature map. The fused feature vector is then processed through a convolutional layer followed by a GMP layer to generate a feature vector, which is then passed through our modified CBAM attention layer to get a more refined and enriched feature vector. The refined feature vector is sequentially passed through multiple convolutional layers, a Global Max Pooling (GMP) layer, and a dropout layer to produce the final feature vector. The final feature vector is then fed to the classifier, which consists of a Dense layer of 196 neurons and a Softmax activation function to give the final predictions.

3 Dataset

The effectiveness of deep learning (DL)-based models in medical image classification is highly dependent on quality, diversity and comprehensiveness of the dataset. In this study, a publicly available dataset from Kaggle [1] was utilized specially curated for the automated detection and classification of common retina disease using AI. The dataset consists of 4217 high-resolution retina images which were originally collected from real world clinical sources and subsequently shared on the Kaggle platform. These images are meticulously annotated and categorized into four distinct disease classes namely, Cataract, Diabetic retinopathy, Glaucoma, Normal (no ocular disease). Each disease class is organized into a separate folder, with the corresponding images inside. To improve the generalizability of the models and mitigate overfitting various data augmentation techniques are applied these technique were especially in cases where class imbalance was observed. In general, this data set serves as a valuable resource for the development and evaluation of robust multiclass classification models for the detection of retinal disease. The structured labeling of images across four categories provides a strong foundation for building AI powered tools for early

Eye disease classification using Feature Fusion based Deep models

7

screening, clinical decision making and large scale ophthalmic diagnostics. Here in the following (Fig.5) are some sample images from each of the classes from our used dataset

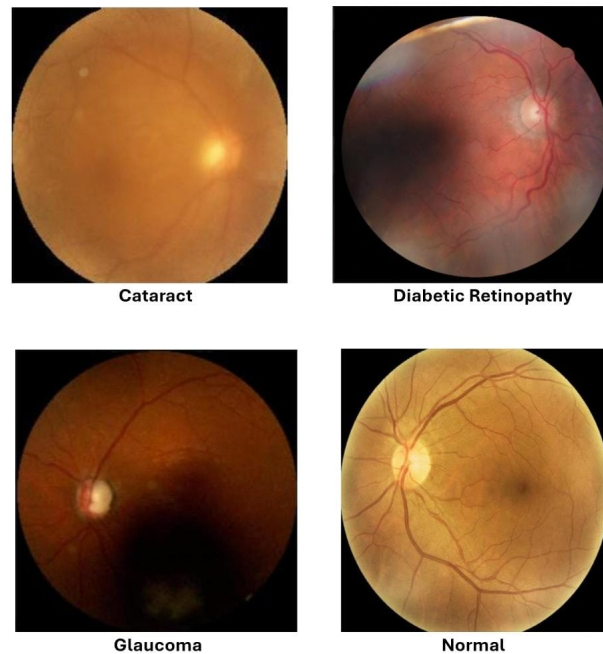


Fig. 5. Sample images from the dataset

4 Result and Discussions

A total of 4217 retina funds images were compiled from publicly available dataset. The experiments have been conducted on a GPU system with the following specifications: Operation System : 64-bit Windows 11 CPU: Intel Core i5 Memory : 16 GB RAM GPU : NVIDIA GeForce RTX 3050

For implementation, Python has been used within the Jupyter Lab interface utilizing popular Deep Learning framework TensorFlow and Scikit-learn, and libraries like NumPy, Pandas and Matplotlib were employed for numerical computations, data manipulation and visualization. The system's GPU was particularly leveraged for accelerating Deep Learning model training, while CPU and memory facilitated efficient data processing and model evaluation. To boost the efficiency and reliability of our deep learning model, various experiments were performed, we took a close look at several pretrained architectures such as InceptionV2, InceptionV3, MobileNetV2, EfficientNetB2 and Xception. The

8 Majumdar et al.

proposed model MobileNetV2 and Xception with CBAM performed better than the other baseline techniques as shown in Table 1.

Table 1. Model Accuracy Comparison

Models	Accuracy (%)
Inception V2	85.78
Inception V3	90.82
EfficientNetB2	90.33
Xception	91.66
MobileNetV2	95.23
MobileNetV2-Xception with CBAM	98.79

MobileNetV2 and Xception came out on top, proving to be the best fit for our final implementation. The experiment was conducted using multi-class ocular disease dataset containing 4 classes: cataract, diabetes retinopathy, glaucoma and normal. All input images were resize to 400×400 pixels. data augmentation technique such as rotation, Zooming, shifting and flipping were employed to enhance generalization. The proposed model combines MobileNetV2 and Xception architecture with CBAM attention mechanism, leveraging transfer learning with pretrained ImageNet weights. The model performance is evaluated using standard classification metrics, confusion matrix visualization and dimensionality reduction techniques.

The model was trained over 100 epochs using a batch size of 16 and optimized with the Adam optimizer (learning rate = $1e-4$). Early stop and reduction of the learning rate were used to prevent overfitting and facilitate convergence. The model demonstrated robust classification performance across multiple categories of ocular diseases. Table 3 provides a summary of the classification metrics: accuracy, recall and F1 score per disease class. Among all categories, glaucoma and cataract was highly detected with the highest accuracy and recall, indicating the model's heightened sensitivity to this condition. These results confirm the model's ability to generalize across a variety of ocular conditions with particularly strong performance in detection cataract and glaucoma. The tabular representation of class wise precision, recall and F1 score is shown in Table2.

Table 2. Class-wise performance metrics of our proposed model

Class (Disease)	Precision	Recall	F1-Score	Support
Cataract	0.997	1.000	0.998	1038
Diabetic Retinopathy	1.000	0.959	0.979	1098
Glaucoma	0.998	0.995	0.996	1007
Normal	0.958	0.998	0.978	1074

Eye disease classification using Feature Fusion based Deep models

9

4.1 Visualization and performance interpretation

To better understand the model's ability to differentiate between ocular disease classes a visualization techniques were used named as t-SNE plots. As shown in Fig 6. t-SNE was applied to the model's predicted probabilities, projecting them into 2d space, the resulting clusters, Coloured by true class labels showed clear separation among disease categorized indicating effective feature learning and minimal overlap highlighting the model's strong discriminative capacity.

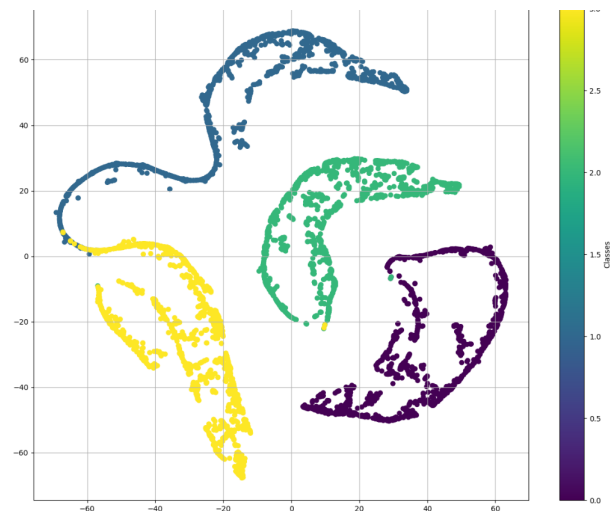


Fig. 6. t-SNE visualization of Predicted Probability

4.2 The confusion metrics and classification report analysis

To assess the classification performance of the proposed hybrid model in detail, a confusion matrix and classification report were computed on the test set consisting of 4217 images distributed. Table 4 present the precision, recall, and F1-score for each class. The model achieved an overall accuracy 98.79% indicating high effectiveness in classifying ocular disease categories

These results demonstrate that the model performs consistently across all classes(disease) with precisions and recall values nearing unity. class 0 (cataract) achieve the highest F1-score(0.9986), while class 1 (diabetic retinopathy), though slightly lower in recall (0.9599) still achieved a high precious (1.0000) indicating minimal false positives. The confusion matrix visualized in Fig 7. confirms the classification strength of the model .most prediction are concentrated along the diagonal ,signifying correct prediction while misclassification are rare and sparsely distributed Such high performance highlights the effectiveness of combining MobileNetV2 and xception features in dual- stream architecture, enhance by a robust training pipeline and optimization strategy.

10 Majumdar et al.

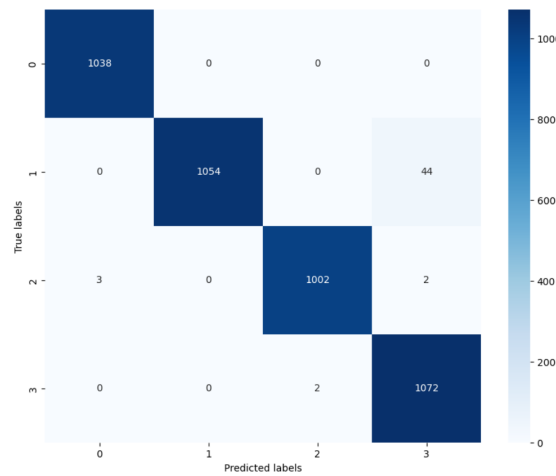


Fig. 7. Confusion Matrix: True Vs Predicted Level

4.3 Comparative analysis

The comparative analysis of the eye datasets [1] are summarized in Table 3. On the test set, the proposed model achieves an accuracy of 98.79%, whereas Sharma et al. [15] achieve a 96.04% accuracy using a Deep transfer learning model based on EfficientNetB5 approach on the eye dataset.

Table 3. Comparative results on eye datasets

Work	Method	Accuracy(%)
Sharma et al. [15]	Deep transfer learning model based on EfficientNetB5	96.04
Prasad et al. [10]	CNN	80
Ramanathan et al. [11]	SVM	90
Ours	Feature Fusion of MobileNetV2-Xception with CBAM	98.79

5 Conclusion

This research proposed a hybrid deep learning model that combines MobileNetV2 and Xception architecture, further enhance with added Convolutional block Attention Module (CBAM), for the automated classification of retina fundus images into four disease categories: Cataract, Diabetic retinopathy, glaucoma and normal. The model leveraged transfer learning with ImageNet weights and incorporated extensive data augmentation techniques to improve generalization and mitigate overfitting. Our proposed model evaluated on publicly available dataset comprising 4217 high-resolution images, the proposed model archived an impressive classification accuracy of 98.79. These results underscore the potential of the

Eye disease classification using Feature Fusion based Deep models

11

proposed model as reliable and interpretable tools for early stage detection of ocular disease. The findings of this study highlight the model's applicability in clinical decision support system and ophthalmology platforms, offering a promise direction for AI-assisted vision screening and preventive eye care.

This research sets the stage for the introduction of AI-driven diagnostic tools into everyday medical practice. Looking ahead, we plan to boost our model's performance by using larger and more varied datasets, which will include images from different demographics and those with rare retinal conditions. In addition, by incorporating explainable AI techniques, we can build clinical trust by shedding light on how the model makes its decisions. There is also a great opportunity to expand this platform into a mobile app, making early eye disease screenings more accessible, especially in remote or underserved areas. With real-time analytics, user feedback loops and integration with electronic health records (EHRs), Vision Health Monitor could evolve into a powerful teleophthalmology tool, playing a vital role in preventive eye care.

References

1. Guna Venkat Doddi. BWorld Robot Control Software. 2025. [Online; accessed 15-April-2025].
2. Eric Topol. *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*. Basic Books, Inc., USA, 1st edition, 2019.
3. Mariusz Bojarski, Davide Del Testa, Daniel Dworakowski, Bernhard Firner, Beat Flepp, Praseoon Goyal, Lawrence D. Jackel, Mathew Monfort, Urs Muller, Jiakai Zhang, Xin Zhang, Jake Zhao, and Karol Zieba. End to end learning for self-driving cars. *CoRR*, abs/1604.07316, 2016.
4. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017.
5. Ajay Agrawal, Joshua Gans, and Avi Goldfarb. *Prediction Machines: The Simple Economics of Artificial Intelligence*. Harvard Business Review Press, Boston, MA, USA, 2018.
6. Yeisson Rivero-Moreno, Sophia Echevarria, Carlos Esteban Vidal Valderrama, Luigi Stefano-Pianetti, Jesus Cordova-Guilarte, Jhon Navarro Gonzalez, Jessica Acevedo-Rodriguez, Gabriela Dorado-Avila, Luisa Osorio-Romero, Carmen Chavez-Campos, and Kathryn Acero-Alvarracin. Robotic surgery: A comprehensive review of the literature and current trends. *Cureus*, 15, 07 2023.
7. Kampanart Huanbutta, Kanokporn Burapapadh, Pakorn Kraissit, Pornsak Sriamornsak, Thittaporn Ganokratanaa, Kittipat Suwanpitak, and Tanikan Sangnim. Artificial intelligence-driven pharmaceutical industry: A paradigm shift in drug discovery, formulation development, manufacturing, quality control, and post-market surveillance. *European Journal of Pharmaceutical Sciences*, 203:106938, 2024.
8. Mohammed S. Alshuhri, Sada Ghalib Al-Musawi, Ameen Abdulhasan Al-Alwany, Herlina Uinarni, Irodakhon Rasulova, Paul Rodrigues, Adnan Taan Alkhafaji, Asim Muhammed Alshanberi, Ahmed Hussien Alawadi, and Ali Hashim Abbas.

12 Majumdar **et al.**

- Artificial intelligence in cancer diagnosis: Opportunities and challenges. *Pathology - Research and Practice*, 253:154996, 2024.
9. Goli Arji, Leila Erfannia, Samira alirezai, and Morteza Hemmat. A systematic literature review and analysis of deep learning algorithms in mental disorders. *Informatics in Medicine Unlocked*, 40:101284, 2023.
10. Krishna Prasad, P S Sajith, M Neema, Lakshmi Madhu, and P N Priya. Multiple eye disease detection using deep neural network. In *TENCON 2019 - 2019 IEEE Region 10 Conference (TENCON)*, pages 2148–2153, 2019.
11. Gauri Ramanathan, Diya Chakrabarti, Aarti Patil, Sakshi Rishipathak, and Shubhangi Kharche. Eye disease detection using machine learning. In *2021 2nd Global Conference for Advancement in Technology (GCAT)*, pages 1–5, 2021.
12. D Shamia, Shajin Prince, and D Bini. An online platform for early eye disease detection using deep convolutional neural networks. In *2022 6th International Conference on Devices, Circuits and Systems (ICDCS)*, pages 388–392, 2022.
13. Ahmed Al Marouf, Md Mozaharul Mottalib, Reda Alhaji, Jon Rokne, and Omar Jafarullah. An efficient approach to predict eye diseases from symptoms using machine learning and ranker-based feature selection methods. *Bioengineering*, 10(1):25, 2023.
14. Abdul Rahaman Wahab Sait. Artificial intelligence-driven eye disease classification model. *Applied Sciences*, 13(20):11437, 2023.
15. Himanshu Sharma, Javed Wasim, and Pankaj Sharma. An efficient system for identification of eye disease in fundus images using a deep transfer learning-based pre-trained model. *Engineering, Technology amp; Applied Science Research*, 14(5):17398–17404, Oct. 2024.
16. S Rubin Bose and V Sathiesh Kumar. Efficient inception v2 based deep convolutional neural network for real-time hand action recognition. *IET Image Processing*, 14(4):688–696, 2020.
17. Jatin Sharma. Enhanced disease detection in corn and maize using efficientnetb2. In *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*, pages 1687–1691. IEEE, 2024.
18. François Chollet. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1251–1258, 2017.
19. Xiaoling Xia, Cui Xu, and Bing Nan. Inception-v3 for flower classification. In *2017 2nd international conference on image, vision and computing (ICIVC)*, pages 783–787. IEEE, 2017.
20. Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4510–4520, 2018.